Chapter 4: Continuous Random Variables and Probability Distributions

PART C:

- NORMAL DISTRIBUTION

Learning Objectives:

At the end of this chapter, the students would be able to:

- ✓ Understand the concept of a continuous random variable and probability distribution
- ✓ Be able to compute the expected value, variance, and standard deviation for a continuous random variable
- ✓ Be able to compute probabilities using a Uniform distribution, Normal distribution, Normal approximation to the Binomial distribution and Normal approximation to the Poisson distribution
- ✓ Continuous Probability Distributions (Probability Mass Functions, Mathematical Expectations)
- ✓ Special Continuous Probability Distributions (Uniform Normal; Normal Approximation to the Binomial Distribution; Normal Approximation to the Poisson Distribution)

4.2 Special CRV: Normal Distribution

- Closely approximates the probability distribution of a wide range of random variables, such as the:

✓Heights and weights of individuals

✓Employee performance scores

✓Sales data

- Serves as the foundation of statistical inference.

4.2 Special CRV: Normal Distribution

A continuous random variable X with probability density function

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad \text{for} \quad -\infty < x < \infty$$

is said to have a normal distribution with parameters μ and σ^2.

Where;

$$\pi = 3.14159 \ldots \quad and\ e = 2.71828 \ldots$$

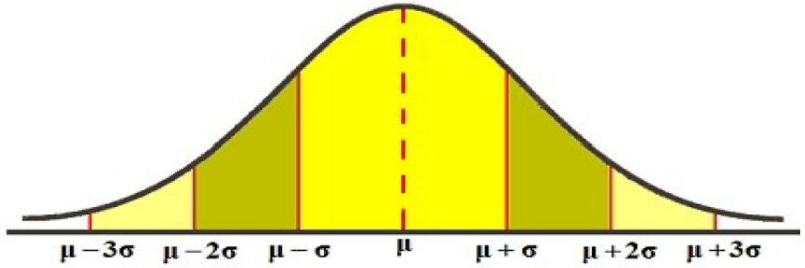
4.2 Special CRV: Normal Distribution

Characteristics of the **Normal Distribution:**

- **Bell-shaped** and **Symmetric** about its mean

Mean = Median = Mode

- **Asymptotic** - that is, the tails get closer and closer to the horizontal axis, but never touch it.



4.2 Special CRV: Normal Distribution

Characteristics of the **Normal Distribution:**

- The normal distribution is **completely described by two parameters**: μ and σ^2. μ is the population mean which describes the central location of the distribution.

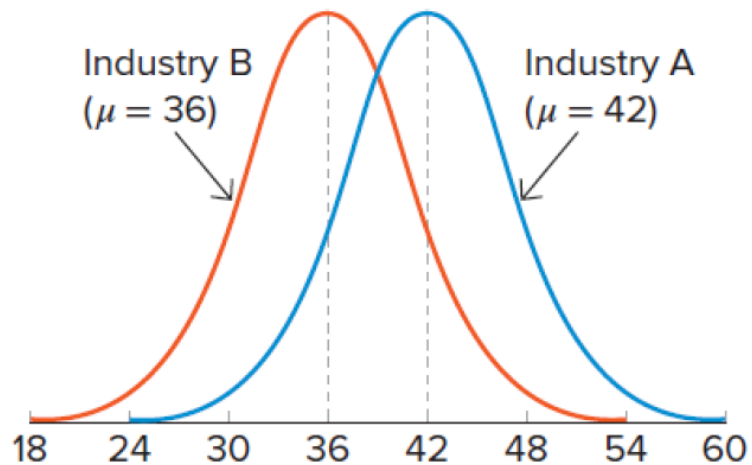
- σ^2 is the population variance which describes the dispersion of the distribution

Example: Suppose the ages of employees in Industries A, B, and C are normally distributed.

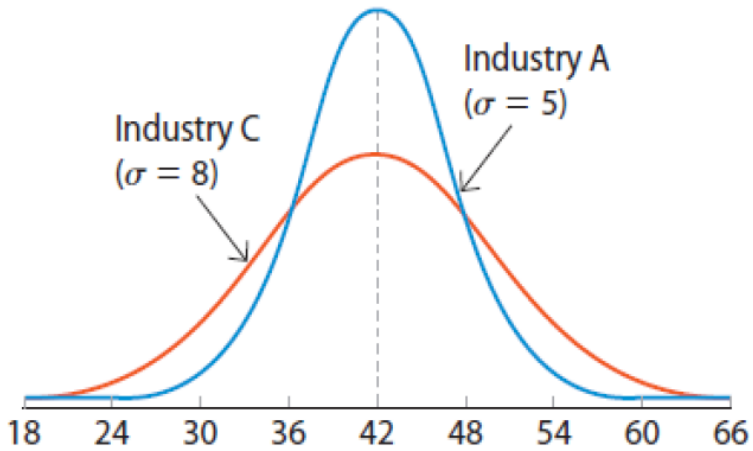
✓ The relevant parameters are:

Industry A	Industry B	Industry C
$\mu = 42$ years	$\mu = 36$ years	$\mu = 42$ years
$\sigma = 5$ years	$\sigma = 5$ years	$\sigma = 8$ years

✓ Compare industries using the Normal curves:



σ is the same, μ is different.



μ is the same, σ is different.

4.2 Special CRV: Normal Distribution

Characteristics of the **Normal Distribution:**

- The normal distribution is **completely described by two parameters**: μ and σ^2. μ is the population mean which describes the central location of the distribution.

- σ^2 is the population variance which describes the dispersion of the distribution

Expected value, Variance, and Standard deviation

If the continuous random variable X is normally distributed with probability density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad \text{for} \quad -\infty < x < \infty$$

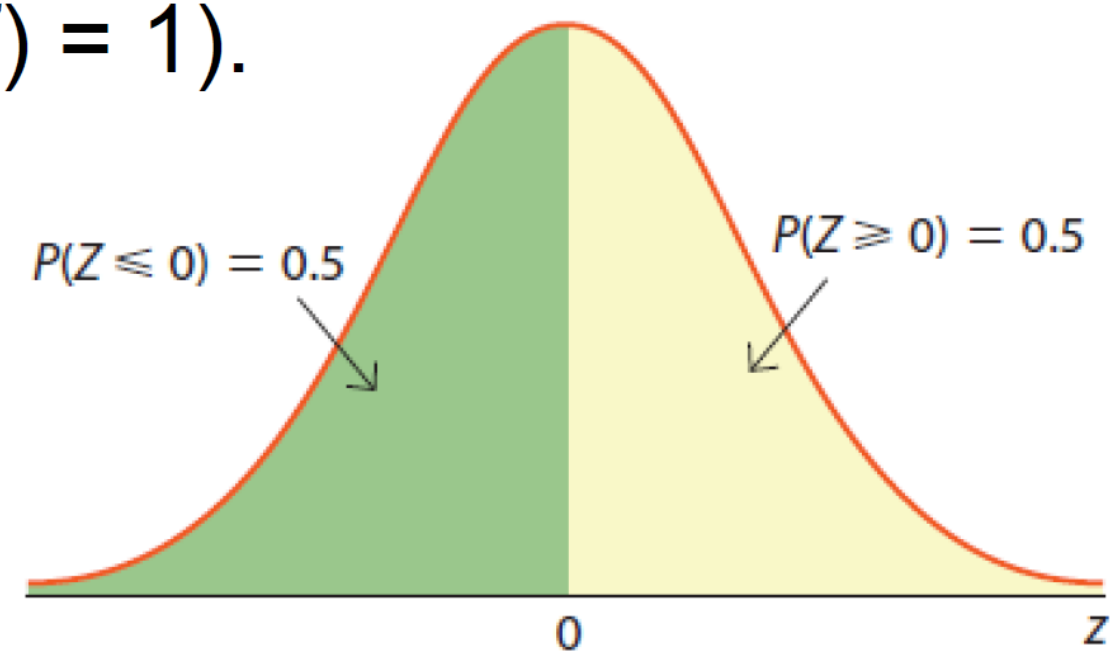
then $E[X] = \mu$ and $\text{Var}[X] = \sigma^2$

In summary, if the continuous random variable is normally distributed with mean=μ and variance = σ^2 , we write $X \sim N(\mu, \sigma^2)$.

The **Standard Normal (*z*) Distribution**.

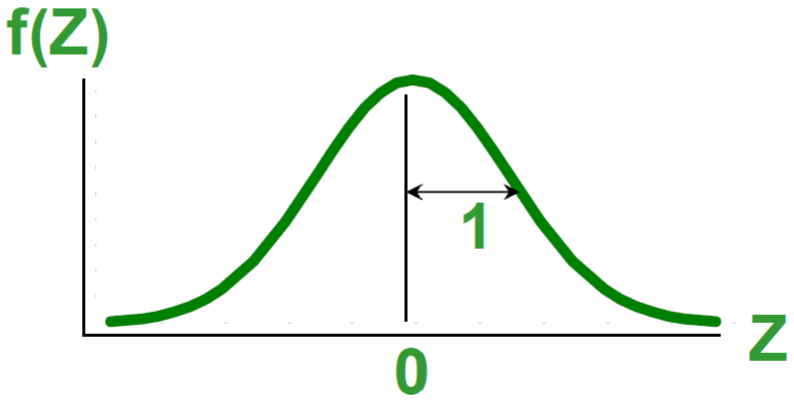
✓ A special case of the normal distribution:

- Mean (μ) is equal to zero ($E(Z) = 0$).
- Standard deviation (σ) is equal to one ($SD(Z) = 1$).



Standard Normal (z) Distribution

- Also known as the “Z” distribution.
- Mean is 0.
- Standard Deviation is 1.



Values **above** the mean have **positive** Z-values.
Values **below** the mean have **negative** Z-values.

Standard Normal (z) Distribution

The standard normal distribution is a special case of the normal distribution. It is the distribution that occurs when a normal random variable has a mean of zero and a standard deviation of 1.

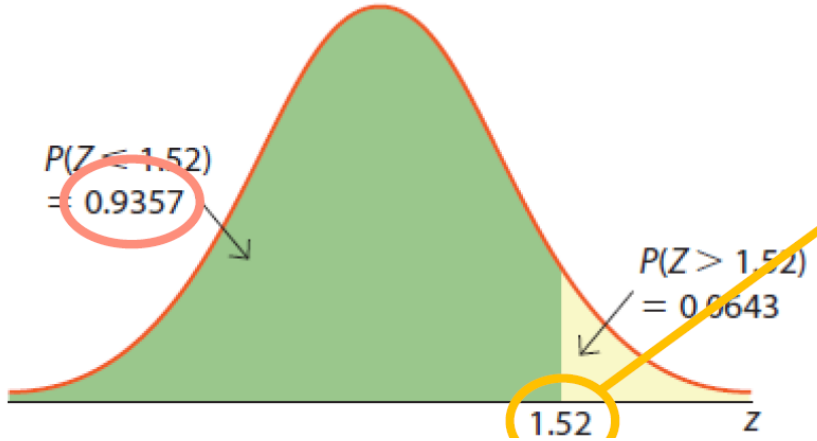
$$Z \sim N(0,1)$$

Every normal random variable X can be transformed into a random variable Z via the following equation:

$$Z = \frac{X - \mu}{\sigma}$$

How-to Find Probability of a given z-value

- ✓ The *z* table provides cumulative probabilities $P(Z \leq z)$ for a given *z*.
- ✓ When *z* is positive, we can look up this probability from the right-hand page of the *z* table.



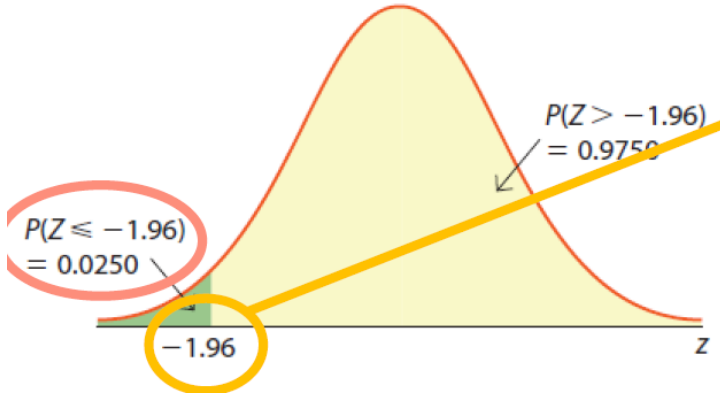
Portion of right-hand page of *z* table.
If *z* = 1.52, then look up

z	0.00	0.01	0.02
0.0	0.5000	0.5040	↓
0.1	0.5398	0.5438	↓
⋮	⋮	⋮	⋮
1.5	→	→	0.9357

How-to Find Probability of a given z-value

- ✓ Remember that the *z* table provides cumulative probabilities $P(Z \leq z)$ for a given *z*.
- ✓ When *z* is negative, we can look up this probability from the left-hand page of the *z* table.

Portion of left-hand page of *z* Table.
If *z* = -1.96, then look up



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06
−3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	↓
−3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	↓
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
−1.9	→	→	→	→	→	→	0.0250

4.2 Special CRV: Normal Distribution

Example 1 (from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 91 – 92)

i. $P(Z > 1.35)$
ii. $P(Z < -1.35)$
iii. $P(Z < 2.47)$
iv. $P(Z > -1.85)$
v. $P(-1.56 < Z < 2.34)$
vi. $P(1.45 < Z < 2.56)$

4.2 Special CRV: Normal Distribution

The **Normal Transformation**

- ✓ Any normally distributed random variable X with mean μ and standard deviation σ can be transformed into the standard normal random variable Z as:

$$Z = \frac{X - \mu}{\sigma}$$ with corresponding values $$z = \frac{x - \mu}{\sigma}$$

By construction: $E(Z) = 0$ and $SD(Z) = 1$.

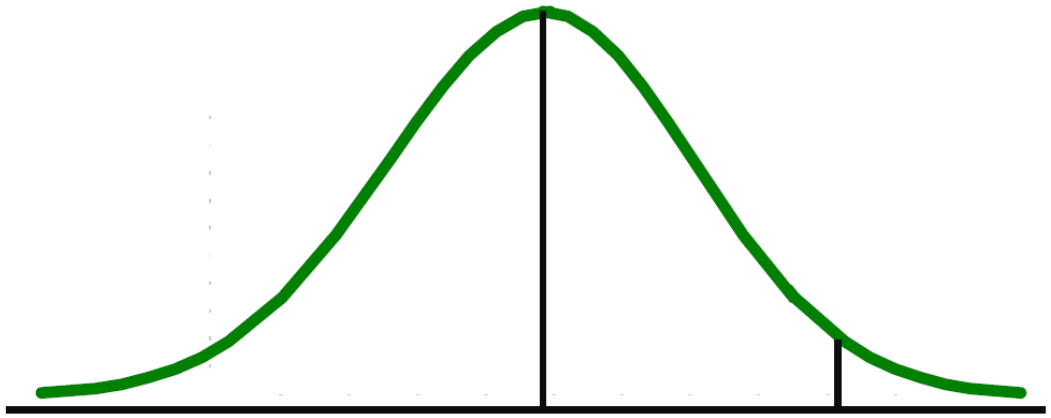
4.2 Special CRV: Normal Distribution

- If X is distributed normally with **mean of 100** and **standard deviation of 50**, the Z value for **X = 200** is:

$$Z = \frac{X - \mu}{\sigma} = \frac{\$200 - \$100}{\$50} = 2.0$$

- This says that **X = 200** is two standard deviations (2 increments of 50 units) above the mean of 100.

4.2 Special CRV: Normal Distribution



100 200 **X** ($\mu = 100, \sigma = 50$)

0 2.0 **Z** ($\mu = 0, \sigma = 1$)

Note that the shape of the distribution is the same, only the scale has changed. We can express the problem in the original units of X or in standardized units (Z).

4.2 Special CRV: Normal Distribution

Example 2 (from textbook Introduction to Probability and Statistics, UUM, 2021); pg. 93)

The mean score for the school and standard deviation were 476 and 107 respectively. Assume that the scores are normally distributed. What is the probability that receiving:

(i) A score from the school falls between 476 and 650?
(ii) A score exceeds 750?
(iii) A score of 540 or less?

4.2 Special CRV: Normal Distribution

Example 3 (from textbook Introduction to Probability and Statistics, UUM, 2021); pg. 95)

A vending machine is regulated so that it discharges an average of 200 milliliters per cup. Assume the distribution of number of drinks is to be normally distributed with a standard deviation equal to 15 milliliters.

(i) What probability of the cups will contain fewer than 225 milliliters?
(ii) How many cups will probably overflow if 233 milliliters cups are used for the next 1000 drinks?
(iii) The probability of the number of drinks is less than k is 0.25, find the value of k.
(iv) The probability of the number of drinks is less than k is 0.0793, find the value of k.